TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	85.5 ± 3.07
Correlation Stability	90.7 ± 0.78
Distribution Consistency	87.8 ± 4.12
Entropy Preservation	90.0 ± 1.79
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.3 ± 0.64
Data Density	100.0 ± 0.00
Coverage Score	97.1 ± 1.30
Feature Accuracy	95.8 ± 1.08
Novelty Score	0.8 ± 0.02

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Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	42.371	+1.171	2.8%	OK
Sex	1.103	1.089	-0.015	1.3%	OK
Steroid	1.506	1.427	-0.079	5.2%	OK
Antivirals	1.845	1.806	-0.039	2.1%	OK
Fatigue	1.351	1.242	-0.109	8.0%	OK
Malaise	1.604	1.508	-0.096	6.0%	OK
Anorexia	1.792	1.685	-0.107	6.0%	OK
Liver Big	1.828	1.839	+0.011	0.6%	OK
Liver Firm	1.583	1.645	+0.062	3.9%	OK
Spleen Palpable	1.800	1.798	-0.002	0.1%	OK
Spiders	1.660	1.685	+0.025	1.5%	OK
Ascites	1.867	1.927	+0.061	3.3%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	85.5 ± 3.07
Correlation Stability (*2)	90.7 ± 0.78
Distribution Consistency (*3)	87.8 ± 4.12
Entropy Preservation (*4)	90.0 ± 1.79
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.3 ± 0.64

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	97.1 ± 1.30
Feature Accuracy (*10)	95.8 ± 1.08
Novelty Score (*11)	0.8 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	0.8064516129032258	0.8064516129032258	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.8387096774193548	0.8419354838709676	0.0032258064516129	
F1	0.5718392134181608	0.518427960927961	0.531016761016761	-0.04082245240139977	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8774193548387098	0.8451612903225806	0.8483870967741934	-0.02903225806451617	
F1	0.6575179722238544	0.5376984126984127	0.5559084774874249	-0.1016094947364297	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	0.7903225806451613	0.7870967741935484	0.00645161290322582	
F1	N/A	0.04895104895104895	0.01818181818181818	0.01818181818181818	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7967741935483871	0.8064516129032258	0.8290322580645161	0.032258064516129004	
F1	0.3846675546675547	0.3116272616272616	0.5062101787101787	0.12154262404262406	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.7967741935483871	0.7838709677419355	0.7806451612903226	-0.01612903225806457	
F1	0.4564257964257964	0.4508336108336108	0.4794263742793154	0.02300057785351905	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.8064516129032258	0.8451612903225806	-0.00645161290322583	
F1	0.5881622725100986	0.4861904761904762	0.5564161328867211	-0.03174613962337748	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.7580645161290323	0.8161290322580644	0.02580645161290321	
F1	0.561449871078354	0.534865680415216	0.584352582402118	0.022902711323763963	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.8354838709677418	0.8451612903225806	0.00645161290322579	
F1	0.5132478632478632	0.5416259230965114	0.5536813186813188	0.04043345543345543	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.8258064516129032	0.8225806451612903	-0.02903225806451617	
F1	0.5385593354014405	0.5007693287105052	0.49555532187111134	-0.04304013530329316	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	51.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	24.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	20.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	57.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	30.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
6	52.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	32.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
3	38.0	1.0	1.0	2.0	1.0	1.0	1.0	1.0	1.0	2.0
4	47.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	1.0	2.0
5	54.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	22.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	26.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
3	52.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
4	47.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
5	52.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
6	25.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	47.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
2	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	64.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	1.0	2.0
4	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
5	52.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	1.0
6	24.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0	1.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	48.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
2	41.0	2.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0
3	50.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
4	23.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
5	51.0	1.0	2.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
6	45.0	2.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
2	57.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
3	61.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	52.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
6	28.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
2	54.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	1.0
3	45.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	39.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
5	40.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
6	43.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	45.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
3	25.0	2.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0
4	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	57.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	28.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	39.0	1.0	1.0	2.0	1.0	1.0	1.0	1.0	1.0	2.0
2	34.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	43.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
4	52.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
5	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
6	32.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	34.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	1.0	2.0
2	37.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
3	27.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
4	27.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
5	56.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	40.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0